HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Actuarial Mathematics and Statistics

F70LA

LIFE INSURANCE MATHEMATICS A

IN-PERSON EXAMINATION

December Examination and Assessment Diet 2023

Duration: 2 hours

Answer all 5 questions.

Total marks: 60

Supplied: Copy of the yellow Formulae and Tables for Actuarial Examinations
Calculator Permitted: Yes

Ethical Commitment: By submitting answers to this paper, I declare that these answers are entirely my own work and have not been shared, in part or in whole or in any form, with any other student, or disseminated in any way.

I also confirm that I have reviewed and understand all University regulations, policies and procedures on assessment misconduct, including but not limited to accessing unauthorised materials during this examination or the use of unauthorised electronic devices.

I understand that infringing this statement would represent an academic misconduct offence subject to disciplinary action according to the University Regulations and Policy regarding Academic Misconduct, with possible significant consequences for degree progression and final degree outcome.

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Question 1

For integer values of x between 25 and 50 inclusive,

$p_x = (1250 – 10x) /1000$
$p_{[x]} = (p_x)^2$
$p_{[x]+1} = (p_{x+1})^{1.5}$
$p_{[x]+2} = p_{x+2}$

Assume a constant force of interest, δ=.1 per year.

(a) How long is the select period?
[1 mark]

(b) Calculate q_{25} and explain whether it's reasonable for human mortality.
[2 marks]

(c) Explain why this selection indicates better or worse health. [2 marks]

(d) Calculate ${}_3p_{[40]+1}$. [2 marks]

(e) Calculate the expected present value of a term assurance for an individual that didn't satisfy the selection criteria, exact age 35, with 2 years of insurance coverage and a sum assured payable at the end of the year of death of £100,000
[3 marks]

(f) Calculate the variance of the present value random variable in (e). [3 marks]

(g) Without doing any more calculations, explain why your answer in (e) would be higher or lower if a higher rate of interest were used for all ages and durations.
[1 mark]

[Total 14 marks]

[**PLEASE TURN OVER]**

Question 2

On 1/1/2000, a life office issued a 25-year non-profit endowment insurance to a life then aged 35 exact. The sum assured of £200,000 is payable at the end of the year of death, or at maturity. Level annual premiums are paid for 10 years.

On 1/1/2010, the policyholder requests to change the policy to pay £Y at the moment of death or age 65 if still alive.

Basis for all calculations:
AM92 select, 6% annual interest. There is an alteration charge of £150, but the basis assumes no other expenses.

(a) Write the equation of value using international actuarial notation and calculate the premium that will be paid on 1/1/2000. Explain why this a net or a gross premium. [8 marks]

(b) Calculate the policy value on 1/1/2010. [2 marks]

(c) Write the equation of value using international actuarial notation and calculate Y. State the required assumption(s), if any.
[6 marks]

(d) Now, assume this basis allows for profit to the insurance company, meaning it's elements are different than what is supported by current observations. Describe how one of the basis elements must compare (>, <, =) with current observations to generate profits for the company.
[2 marks]

[Total 18 marks]

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Question 3

An insurance company sells a whole life annuity-due to a policyholder aged 70 exact that provides annual payments of 10,000.

The interest is assumed to be 6% pa effective and mortality is assumed to follow the AM92 Select table.

Let Z denote the present value random variable for this annuity.

(a) Define Z as a function of a future lifetime random variable. Use appropriate subscripts and/or superscripts. [3 marks]

(b) Calculate Var (Z) [3 marks]

(c) Calculate Pr(Z ≥ 88,888). [3 marks]

(d) Explain how Var (Y) would compare (>,<,=) to Var(Z) where the only difference with Y is that it's a whole life immediate annuity. [2 marks]

[Total 11 marks]

Question 4

Using the stated assumption between integer ages and ELT15 (Females), calculate:

(a) ${}_{2.3|4}q_{70.3}$ uniform distribution of deaths (UDD) [3 marks]

(b) Explain the meaning of the notation used in (a) [1 mark]

(c) ${}_{14.3}p_{40.7}$ constant force of mortality [3 marks]

(d) ${}_{0.2}q_{40}$, ${}_{0.2}p_{40}$, $\mu_{40.2}$ UDD [3 marks]

[Total 10 marks]

[**PLEASE TURN OVER]**

Question 5

On 1 January 2000, a company issued 100 identical whole-of-life assurance policies to women who were then aged exactly 60.

Each policy has a death benefit of £10,000 that will be paid at the end of the year of death. Each policy was purchased by a single premium.

The basis for calculating all policy values is as follows:

- Mortality (limited amount shared): $q_{74} = .01$, $q_{75} = .03$, $q_{76} = .04$,
- Interest: 6% per year
- Expenses: 1% of each death benefit payment
- You are also told that policy value, V(14) = 5500

On 1 January 2015, 80 of these contracts were still in force. During 2015, the office's experience was as follows:

Interest was earned at 6% per year.

Two policyholders died.

Expenses of £240 were incurred on 1 January 2015.

The company had no expenses at the time of claim payments.

(a) Calculate the amount that would be held in a bank to satisfy the policy value requirements for a single policy on 1/1/2015. [3 marks]

(b) State whether mortality profit in 2015 is (>, <, or =) 0. Explain why. [2 marks]

(c) State whether interest & expenses combined in 2015 result in profit (>, <, or =) 0. Explain why. [2 marks]

[Total 7 marks]

END OF PAPER